A Machine Learning Approach to Predicting the Oscars

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***Abstract*— The Academy Awards, or Oscars, are the film industry's highest honour. Since debuting in 1929, they have become a global phenomenon, reflecting their importance in entertainment. Employing AI to predict the Oscars is a relatively underexplored topic. This paper investigates the application of machine learning techniques for predicting Academy Award (Oscar) winners in six major categories: Best Picture, Best Director, and the four Acting awards using a dataset spanning 1961-2025 and incorporating both established and newly engineered features. Logistic Regression proved most effective for Best Picture, Best Director and Male Acting categories, with Support Vector Machines performing best for Female Acting categories, suggesting non-linear patterns in how female performances are evaluated. Key predictors included Directors Guild of America (DGA) wins, which contributed most to both Picture and Director outcomes, and a novel Best Editing nomination feature, which strongly factored into Best Picture success. Results showed top pick accuracy of 85% (Picture), 92% (Director), and 69% (Male and Female Acting), meeting or exceeding benchmarks from prior research while addressing methodological flaws. Importantly, gender-specific analysis and modelling revealed distinct factors influencing male and female acting awards, highlighting unconscious bias in Academy voting patterns.**

Keywords*— *Oscars prediction, machine learning; movie industry; Oscars.

I. Introduction

The Academy Awards (the Oscars) are the film industry's most prestigious recognition of cinematic achievement. Since the first awards in 1929, the Oscars have evolved to become a global phenomenon, with the 2025 ceremony reaching roughly 20 million viewers. As well as cultural significance, the Oscars can have significant value with regards to both reputation and financial success for studios, filmmakers and more [1].

The Oscars are voted for by over 10,000 members of the Academy for Motion Picture Arts and Sciences (AMPAS) who are industry professionals across all disciplines, and who vote for the winner in their relative category/categories (with all members voting for Best Picture) [2]. While the voting process aims to capture subjective artistic judgment, patterns in Oscar outcomes suggest there are underlying predictive factors affecting this judgement that manifest differently between categories, and can be identified through statistical analysis.

The influence of the Academy Awards within the entertainment industry is extremely high. In 2001, Nelson and colleagues examined the tangible impact of winning an Oscar on a film's commercial performance, including box-office longevity and distribution reach.

Their analysis showed that receiving a nomination or an award for Best Actor or Best Actress was associated with substantial increases in the number of cinema screens showing the film (by roughly 41% for nominations and over 120% for wins). Similarly, films nominated for or winning Best Picture experienced comparable growth, with venue counts rising by about 85% and 200%, respectively. They also reported that receiving Oscar recognition is closely linked to large financial gains. Specifically, a Best Picture nomination was found to raise box-office revenues by nearly 99% during the six weeks between the nominations and the ceremony. Winning the Best Picture award had an even greater effect, with revenues increasing by roughly 245% in the first month after the awards and about 201% in the following 7 weeks [1].

Given this influence, there have been some attempts to employ machine learning algorithms to predict Oscar winners, although it is still considered a largely underexplored area. Regarding the characteristics of Oscar winners, Jacqueline R. Carlton's 2021 statistical study provides insight into key influencing factors. Her analysis of data from 2000 to 2019 showed that actresses who won Oscars tended to be significantly younger than their male counterparts, suggesting possible gender-related disparities in award outcomes [3].

One prominent contributor to research on Oscar prediction is Iain Pardoe, who first explored the topic in his 2005 paper "Just how predictable are the Oscars?" [4] and later expanded on this work in a 2008 collaboration with Simonton [5]. Both studies employed discrete choice models to generate yearly forecasts for the four major award categories and evaluated their performance using out-of-sample, one-year-ahead prediction errors. The 2008 paper built upon the earlier work by comparing Bayesian and maximum likelihood estimation methods and by examining whether a more advanced mixed logit model offered improvements over the traditional multinomial logit approach. They showed an accuracy of 69% over the four major award categories.

In 2006, David Kaplan attempted to create a logistic regression model that could predict the Best Picture winner only [6]. His data span from 1965 to 2004. He found that the total number of nominations, winning the Golden Globe Award for Best Picture in the Drama category, being an epic

or biographical movie, and having the director win the DGA award positively affect the probability of winning Best Picture.

Machine learning software company BigML have an ongoing blog series where they make predictions for almost every Oscar category each year, including the most recent 2025 ceremony [7]. These predictions are based on a large, public dataset, containing data from 2001 to 2025. The company uses their software to find the best-performing supervised models and combine them based on the phi-coefficient metric, to make their predictions more robust. They reported an accuracy of 68% over 19 categories.

Váradi's research provides the most directly relevant precedent for this study, as a recent comprehensive application of modern machine learning techniques to Oscar prediction across multiple major categories [8]. Using data from 1961 to 2018, Váradi tested three algorithms – Logistic Regression (LR), Random Forest (RF), and Support Vector Machines (SVM) – the same that are employed in this study. Váradi's methodology established several important precedents. His binary classification approach using ranked probabilities provides the framework adopted in this research, where the nominee with the highest predicted probability in each category and year is designated as the predicted winner. His feature engineering approach, incorporating preceding award ceremony results as the primary predictive variables, demonstrates the continued relevance of traditional earlier awards in contemporary prediction models. His RF classifier achieved an AUC-ROC of 0.91, 0.96, 0.88 and 0.78 in Best Picture, Best Director, Best Actor & Actress, and Best Supporting Actor & Actress, respectively. These findings align with the intuitive understanding that certain categories may be more predictable than others, while establishing RF as a particularly effective algorithm for this classification problem. However, Váradi's approach contains flaws in the assessment phase and methodological limitations that this research addresses. Most critically, he reported only two accuracy metrics: sensitivity and ROC scores. The sensitivity scores were "calculated after the models were fit to the entire dataset and not only to the test set" [8]; this makes this metric unreliable. As for the ROC scores, which were calculated on the 30% test dataset, even high ROC scores could be misleading, as they could hide other low scores such as the F1-score, particularly when the dataset is imbalanced. Additionally, Váradi did not distinguish between male and female acting categories in his variable selection or modelling approach, potentially missing important gender-specific patterns in Academy voting behaviour.

The literature reveals some methodological challenges that should be addressed when attempting Oscar prediction research. First, choosing the most impactful variables presents ongoing challenges. While all successful models incorporate preceding award ceremony results as primary predictors, there is less consensus on other features. Pardoe [4] found that variables like genre classification, runtime, and release date did not improve his models, while Kaplan [6] and BigML [7] found these same variables to be statistically significant. These conflicting findings suggest that variable importance may vary by time period, category, or modelling approach.

The literature lacks gender-specific analysis of winners of acting categories. There has been research into gender bias in film industry evaluation and the different criteria by which male and female performances are historically assessed. For example, Doughman and Khreich's 2025 study on bias in movie reviews analyzed more than 17,000 critiques, quantifying both benevolent and hostile forms of sexism [9]. Their results showed that benevolent sexism scores were about 44% higher in films featuring women in major acting, writing, or directing roles, while hostile sexism scores were roughly 149% higher for female-led films compared with male-led ones. The authors concluded that such disparities indicate unconscious bias among critics, which may shape film reception and influence the professional outcomes of women in the industry (factors that can, in turn, affect Oscar results). Separate modelling of male and female acting categories represents an innovation that could improve prediction accuracy while revealing insights about Academy voting patterns.

This study focuses on predicting Oscar winners in the six major categories: Best Picture, Best Director, Best Actor and Actress in a Lead Role, and Best Actor and Actress in a Supporting Role. Using data spanning from 1961 to 2025, this research uses machine learning techniques to identify the most significant predictors of Oscar success, and develop models that can accurately identify winners. This investigation contributes to the field of Oscar prediction through the implementation of feature engineering tailored to each category, and gender-specific modelling for acting categories. It is also a comprehensive comparison of Logistic Regression, Random Forest, and Support Vector Machine algorithms for identifying Oscar winners, a unique problem combining binary classification and ranked probability.

II. Data Preparation

A. Data Acquisition

Three separate datasets were used, all of which included all nominees for a specific category from the 1961 Oscars up until the most recent 2025 awards – one for Best Picture (390 films), one for Best Director (325 nominees), and one which combined all four Acting categories (1,300 nominees across Lead Actor, Lead Actress, Supporting Actor and Supporting Actress). The primary source of the data was Váradi's undergraduate dissertation [8] as well as updated datasets up until 2025 on GitHub [10]. These datasets included features sourced from IMDb [11], Rotten Tomatoes [12] and more, some of which were adapted or removed for this research.

B. Data Cleaning

There were 19 missing Age values in the Acting data, and 15 Acting nominees missing in their entirety (the latter was found after calculating the expected number based on the shifting number of yearly nominees since 1961). These missing values were filled in manually using the relevant ceremony pages on AMPAS' official website and IMDb. Other data cleaning tasks included removing irregular characters from nominee and film names for more efficiency

when scraping data for new features and changing datatypes from floats to integers where appropriate.

C. Feature Engineering

One of the novelties we have tried to incorporate into this research was that different Oscar categories would require different features and approaches for accurate prediction. As such, we engineered several new features tailored to each of the datasets, most of which captured demographic information about nominees, largely based on one of the authors' domain knowledge of the film industry. Two new features were added to the Picture and Director datasets – the runtime of the nominated film (or the film the director or actor was nominated for) in minutes, and a binary column representing whether the United States of America was one of the countries in which the film was financed and produced (note – this doesn't necessarily mean it was filmed in the USA). The latter, and other America-related features, were added to test whether the Academy favours American nominees, as the Oscars are an American awards ceremony. This data was obtained from the community-maintained: The Movie Database (TMDb) [13], using an API scraper. While TMDb provides extensive coverage and generally reliable data, occasional inaccuracies may exist due to its community-driven nature. The 99.8% successful match rate of the scraper and manual spot-checking of sample entries suggested good overall data quality suitable for this analysis.

One new feature was added only to the Best Picture dataset – a binary column showing whether the film also received a Best Editing Oscar nomination that year. This was based on industry knowledge and speculation for years about the link between the Editing and Picture awards. 57% (224 out of 390) of Best Picture nominees from 1961-2025 also have a Best Editing nomination, rising to 89% (57 out of 64) of winners, indicating this feature's potential predictive power.

Five new features were added to the Best Director dataset. Two were binary value columns indicating whether the director had been nominated for or won an Oscar before that year (features that counted their nominations/wins were already present). Three were demographic-related features – their age at that year's ceremony, whether they were born in America, and their gender. The age data was gathered using IMDb's non-commercial datasets. The birthplace and gender data was collected using a Wikidata scraper, with manual data added for 16 directors that were missed.

Two new features were added to the Acting dataset. The first was, as with the directors, whether the nominee was born in America, again gathered using a Wikidata scraper. The second was whether the actors were playing a real-life person in their nominated role and was filled in entirely manually (and therefore carefully verified) through research of each role on Wikipedia, IMDb and more. A role was only considered a real person if it was an actual verifiable real-life figure and had the same name as the person it was based on – it does not include composite characters or those only loosely inspired by real people. The final datasets used in this study are publicly available at https://zenodo.org/records/17494990

III. Modelling and Evaluation

A. Classification vs. probability

This paper aims to predict Oscar winners by generating probabilities for each nominee, rather than simply binary classification. However, implementing true probability modelling would require multinomial logistic regression with a fundamentally different dataset structure – one row per Oscar year containing all nominees and their features, rather than one row per nomination. This would reduce each dataset to only 64 rows (one for each year from 1961 to 2025), which would be insufficient for effective machine learning model training. Instead, this research treats the problem as binary classification, where each model outputs a probability of winning for individual nominees. These probabilities are then ranked within each Oscar year to identify the predicted winner (measured as 'top pick accuracy'). While the probabilities in each year don't sum to 1.0, this approach still effectively addresses the core prediction task of accurately ranking nominees and identifying winners. The evaluation methodology accounts for this approach through the use of comprehensive metrics including top one/two/three pick accuracy and F1 scores, ensuring there's still a robust assessment of each model's predictive performance regardless of probability calibration.

B. GroupShuffleSplit

Before fitting a model to the data, it needed to be split into training and testing subsets (a validation subset was not used due to the datasets being relatively small). To ensure nominees from the same year stayed together in either the training or test sets, and were never split across both, the *GroupShuffleSplit* method was used, with the groups defined using the Oscar_Year column. This also allowed for a random shuffled split with regards to chronology, as having all the training data from early years and all the test data from more recent years would have provided biased results due to potential temporal Oscar trends.

C. Logistic Regression

Logistic Regression was selected as the baseline model due to its simplicity and interpretability, in that it allows for identification of the most predictive factors through in-built feature importance analysis using coefficients.

Firstly, the Best Picture data, with an 80%-20% train-test split, was run through the LR model with default parameters. This provided strong initial results – a top pick accuracy of 85%, and AUC-ROC of 0.9008. Testing with a 70%-30% train-test split, and with the model's class_weight parameter set to 'balanced' – to address the fact there are many more losing nominees than winners – showed no improvement in performance.

A baseline LR model was then run on the Best Director dataset, with results of 92% top pick accuracy and an AUC-ROC of 0.9645 – this quickly indicated that the Director award winner is highly predictable.

Baseline results for the full Acting dataset (both genders and all four categories combined) were 85% top pick accuracy and an AUC-ROC of 0.8081. The overall F1 score was 0.5454

– but metrics for each individual category showed a significant difference across genders. The F1 scores for Lead and Supporting Actor were 0.72 and 0.6154 respectively but dropped to only 0.4615 and 0.3846 for Lead and Supporting Actress. This clearly indicates that a tailored approach is required to improve the Female Acting results, and that the most predictive factors may vary between gendered categories. To investigate this further, the Acting dataset was split by gender – the Lead/Supporting Actor nominees, and those for Lead/Supporting Actress. Running a baseline LR model on the gendered datasets confirmed variation in performance, with a 69% top pick accuracy and overall AUC-ROC of 0.8255 on the male test data, compared to 62% and 0.7500 on the female test data.

D. Feature Analysis

To determine which features – both from the existing dataset and our newly engineered ones – are most predictive of Oscar winners across each category, extensive feature analysis was carried out, removing features in isolation and running Logistic Regression models to evaluate performance. For the Best Picture dataset, results showed that only the new Best Editing Nomination feature had an effect, with evaluation metrics including the AUC-ROC score worsening when it was removed. The new *Runtime* and *Produced In America* features had no real effect.

Testing on the Best Director dataset revealed that all five new features were essentially redundant, with no real effect on results. This suggests that the original features already captured patterns effectively and supports the baseline model results that this is the most predictable category.

For the Acting datasets, results for all splits of data (the whole dataset, just the Male categories, and just the Female categories) showed that the new Real Person feature was consistently irrelevant, and its removal had no impact on performance metrics. This suggests that although biographical films tend to have a strong presence at the Oscars, actors playing an existing historical figure does not contribute to whether they win or not. Where the gender-split data diverges is with the *Born In America* feature – when removed, it had minimal impact on results for the Male Acting data, but a more substantial negative impact for the Female Acting data. This suggests that geographic bias affects the female awards differently than the male ones.

Tests were then carried out removing the three Ratings features (one from IMDb, two from review-aggregator site Rotten Tomatoes – the audience and critic score). This was done because to work effectively on unseen data, the models must be based on data that can be gathered entirely before the Oscar ceremony takes place. These ratings, having been gathered in most cases after the film won or didn't win awards, are not technically pre-ceremony features, and ratings could have been inflated or deflated due to changing critical consensus over time or because of the very fact that a film did or did not win Oscars. These tests showed that the Best Picture and Female Acting models were still fully viable without the ratings, and the Best Director model performed better without them – the Male Acting model was the only one that showed some dependency on ratings, with a 7% drop in top pick accuracy when they were removed. This indicates that three of the four models can achieve strong pre-ceremony prediction, whilst the Male Acting model could shift in performance when predicting for future Oscar ceremonies.

After this comprehensive testing, four best LR models were selected based on performance in top pick accuracy, F1 score, AUC-ROC and more. These four models were validated by using 5-fold cross-validation. The Picture and Director models showed the strongest cross-validation performance, with high AUC-ROC scores and low variance, confirming their top pick accuracies are robust and the models are stable. Male Acting's cross-validation results show reasonable consistency – but the Female Acting model returned a substantially higher variance in F1 scores, reflecting the increased challenge found in predicting these categories.

The optimal feature combinations for each dataset are as follows. The Best Picture model was one that excluded the Produced In America feature. The Best Director model excluded all five new features, plus the universal Runtime and Produced In America features, and the three Ratings. The best Male Acting model was the baseline one, including all features, and the best Female Acting model excluded only the new Real Person feature.

Feature importance using coefficients was calculated for these four models, with the results shown in Figures 3 to 6. This revealed clear patterns in what drives Oscar success across different categories. For Best Picture, the Directors Guild Award (DGA) win was by far the most important predictor, followed by the new Best Editing nomination feature that was added for this study. For Best Director, the DGA win was even more dominant, suggesting these awards are highly predictable once the Directors Guild announces their winner.

The Acting categories showed more complex patterns and notable gender differences. For Male Acting, SAG wins were most important, but previous Oscar wins actually hurt their chances, suggesting the Academy spreads awards around rather than rewarding the same people repeatedly. Age also helped male actors, with older performers having better odds. Female Acting showed different patterns entirely: while SAG wins were still important, they benefited from appearing in war, crime, and biography films, whereas these genres either didn't matter or hurt male actors' chances. Female actors were also negatively affected by films released in the fourth quarter, suggesting different seasonal dynamics for how their performances are evaluated. These gender differences indicate that the Academy may unconsciously apply different criteria when judging male versus female performances.

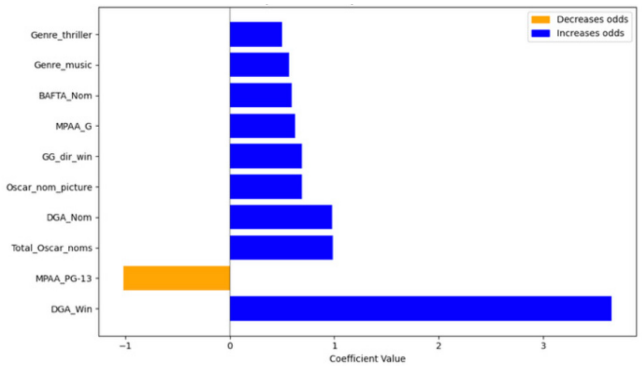


Fig. 3. Top 10 Feature importance for best Picture model

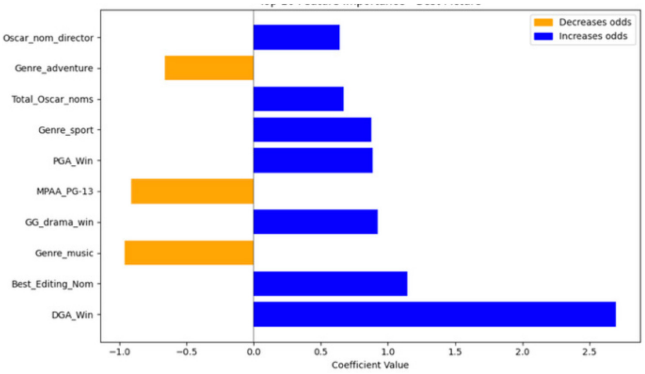


Fig. 4. Top 10 Feature importance for best Director model

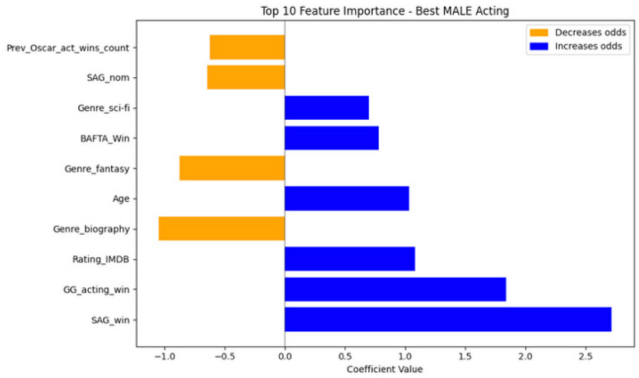


Fig. 5. Feature importance for best Male Acting model

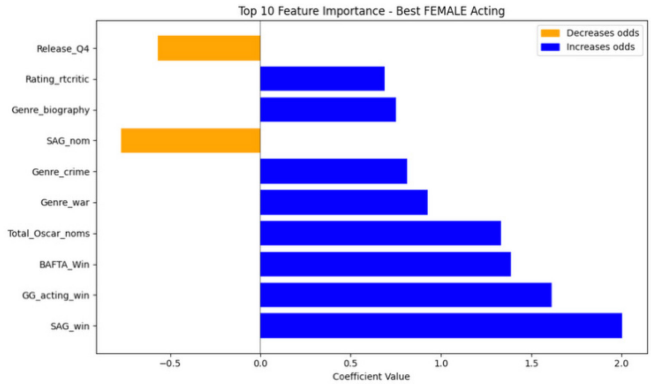


Fig 1. Top 10 Feature importance for best Female Acting logistic regression model

E. Support Vector Machines

To check whether performance could be improved by using different machine learning algorithms, the optimal feature combination for each of the four datasets (Acting being split by gender) was run through a Support Vector Machine (SVM) model, using default parameters.

Logistic Regression clearly outperformed SVM on the Best Picture (85% vs 77% top pick accuracy) and Male Acting (69% vs 62%) data. The Best Director data results were essentially the same, maintaining the top pick accuracy of 92%. The Female Acting dataset was the only one to show meaningful improvement using an SVM model, with top pick accuracy increasing from 62% with LR to 69% with SVM.

This reinforces our core hypotheses that female acting categories require different modelling approaches to male ones, and SVM's ability to capture non-linear relationships appears to better handle the complexity in female acting predictions. As such, this SVM model was chosen as the best performing model for the Female Acting dataset, and cross-validated, showing similar strong (but with higher variance) results as the Logistic Regression version.

F. Random Forest

The optimal feature combination for each dataset was also run through a Random Forest Classifier model, again to test the efficacy of another machine learning algorithm, and because Váradi found it to provide the best results in his study. Random Forest showed signs of severe overfitting on all datasets, with perfect training scores but poor generalization. This is likely due to relatively small sample sizes and high feature dimensionality. As such, it was not considered among the best performing models.

G. Best model selection

The highest performing models for each dataset are outlined in Table 2, with LR providing best results for the Picture, Director and Male Acting data, and SVM for Female Acting.

TABLE II. BEST PERFORMING MODELS

Dataset	**Model**	**Top Pick Accuracy**	**AUC-ROC**	**F1 Score**
Best Picture	LR	85%	0.930 (±0.035)	0.769 (±0.109)
Best Director	LR	92%	0.949 (±0.023)	0.892 (±0.038)
Male Acting	LR	69%	0.847 (±0.031)	Actor: 0.656 (±0.091) Supp Actor: 0.662 (±0.062)
Female Acting	SVM	69%	0.817 (±0.044)	Actress: 0.692 (±0.161) Supp Actress: 0.523 (±0.090)

IV. Discussion and Conlcusion

This study primarily builds upon the foundational dataset and methodological approach by Váradi [8], who demonstrated the effectiveness of machine learning techniques for Oscar prediction on data spanning 1961 to 2018. Váradi's work also used a binary classification with ranked probabilities approach (as this paper does), and established key predictive features including preceding awards ceremonies, total Oscar nominations and genre-based features. Our research introduces several methodological innovations that go beyond that original framework. Most significantly, gender-specific modelling and analysis was implemented for acting categories. New features were also incorporated, including Best Editing nomination for Picture prediction (which proved to be important), while certain features were removed based on results. Although he reported high AUC-ROC, as mentioned earlier, that was the only reliable metric which could be misleading without providing other metrics such as F1 Score, especially in unbalanced datasets.

BigML [13] used the same top pick accuracy method with a result of 68% when predicting the 2025 ceremony winners across 19 categories, using one big dataset (and the same features) for all 19. This lower accuracy suggests prediction becomes more challenging with a wider scope, however, our results demonstrate that strategic grouping, feature engineering and algorithm choice can help maintain good performance across the main six categories (69% – 92%).

Best Picture top pick accuracy of 85% matches Kaplan's [6] comparable 'success rate' of 85-90% (depending on his model specification), achieved by selecting the nominee with the highest predicted probability of winning each year. However, the 85% Picture and 92% Director top pick accuracy this study achieved exceeds Pardoe and Simonton's [5] overall result of 69% when combining the four major categories.

This research successfully demonstrates that machine learning can achieve strong predictive performance for Oscar winners, with accuracy ranging from 69% for Female Acting to 92% for Best Director categories. The study's core hypothesis was validated: different Oscar categories require tailored features and modelling approaches. Most significantly, gender-specific analysis in Acting categories revealed distinct voting patterns, with female actors' chances being more affected by genre and release date, where male actors' odds were more affected by age and previous Oscar wins. This shows not only methodological improvement, but also cultural insight into how systemic biases manifest in awards wins.

The newly engineered Best Editing nomination feature proved highly predictive for Best Picture, and Directors Guild Award wins emerged as the strongest predictor across Picture and Director. Algorithm selection also varied by category, with Logistic Regression optimal for most datasets but Support Vector Machines superior for Female Acting, suggesting non-linear relationships in female performance evaluation.

This study's findings have practical implications for many key stakeholders across the film industry. Accurate Oscar prediction could aid studios and distributors in refining their marketing strategies and production decisions; filmmakers and talent with regard to the projects they pursue; industry analysts and journalists in knowing which films to highlight and which may benefit from being championed early; and moviegoers in deciding what to see in theatres at a time when cinema ticket prices are rising, and streaming films at home is more accessible than ever.

Future work should explore multinomial probability modelling to produce true probability distributions, temporal trend analysis across the decades of the Oscars, and attempt to identify the best approaches for a wider range of categories beyond the main six tackled here. Testing out deep learning models could better capture complex feature interactions, particularly for Female Acting categories, and a real-time prediction system would be useful for incorporating awards season developments as they happen.

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